ESSENTIAL EXAM TOPICS GUIDE

Computer Science / Informatics Practices Quick Revision

1. Computer Networks (Case Study Specialist)

Focus on the 5-mark case study revolving around setting up new networks. Key decision-making skills required:

- **Server Location:** Always suggest the block with the maximum number of computers to reduce traffic and cabling costs.
- **Cable Layout:** Typically **Star Topology** is preferred for connecting buildings to a central server.
- **Networking Devices:**
  - `Switch/Hub`: Used to connect multiple computers within a single department or block.
  - `Repeater`: Used to amplify signals if cable length exceeds 70-100 meters.
- **Network Scale:** Distinguish between **LAN** (Local), **MAN** (Metropolitan), and **WAN** (Wide Area, e.g., different cities like Mumbai and Jaipur).

2. Python Pandas (DataFrames & Series)

Coding proficiency is essential for these core data structures:

- **Creation:** Practice creating DataFrames/Series from dictionaries, lists, or `ndarrays`.
- **Structural Changes:**
  - `df.drop()` to remove rows or columns.
  - `df.rename()` for modifying column headers.
  - Direct assignment for adding new columns.
- **Data Access:** Master **Slicing** and **Filtering** using boolean indexing and condition-based selection.

3. Data Visualization (Matplotlib)

Fill-in-the-Blanks Tip: Learn the sequence: Import → Setup Data → Plot → Label → Show.

- **Importing:** `import matplotlib.pyplot as plt`
- **Plotting:** `plt.plot()` for Lines; `plt.bar()` for Bar Charts.
- **Annotation:** `plt.xlabel()`, `plt.ylabel()`, and `plt.title()`.
- **Display:** `plt.legend()` and `plt.show()`.

4. SQL Database Queries

- **String Functions:** `INSTR()`, `LENGTH()`, `SUBSTRING()`, `LOWER()`, and `UPPER()`.
- **Math Functions:** `MOD()` for remainders and `ROUND()` for precision.
- **Aggregates & Grouping:** `SUM()`, `AVG()`, `MAX()`, `MIN()`, `COUNT()` + `GROUP BY` and `ORDER BY`.
- **DDL Commands:** Writing `CREATE TABLE` with appropriate data types and `PRIMARY KEY` constraints.

5. Societal Impacts & Cyber Ethics

- **E-waste:** Hazards of improper disposal and the necessity of recycling electronics.
- **Intellectual Property Rights:** Understanding Copyrights, Trademarks, and the dangers of **Plagiarism**.
- **Cybersecurity:** Definitions of **Phishing** (credential theft) and **Cyberbullying**.
- **Digital Footprints:** Differences between **Active** (knowingly shared) and **Passive** (collected without user knowledge) footprints.
